

Big Data Frameworks and MLOps

Lab 7 - Airflow and MLflow

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Setting the Airflow Home Directory and Creating the Airflow Workspace

```
hadoop@BDTT: ~  
hadoop@BDTT:~$ export AIRFLOW_HOME=~/.airflow  
hadoop@BDTT:~$ mkdir -p ~/.airflow  
hadoop@BDTT:~$ echo $AIRFLOW_HOME  
/home/hadoop/airflow  
hadoop@BDTT:~$
```

Initializing and Migrating the Airflow Metadata Database

```
hadoop@BDTT:~$ python3 -m airflow db migrate  
2026-03-13T02:44:00.436674Z [info      ] setup plugin alembic.autogenerate.schema  
s [alembic.runtime.plugins] loc=plugins.py:37  
2026-03-13T02:44:00.436947Z [info      ] setup plugin alembic.autogenerate.tables  
[alembic.runtime.plugins] loc=plugins.py:37  
2026-03-13T02:44:00.437257Z [info      ] setup plugin alembic.autogenerate.types  
[alembic.runtime.plugins] loc=plugins.py:37  
2026-03-13T02:44:00.437395Z [info      ] setup plugin alembic.autogenerate.constr  
aints [alembic.runtime.plugins] loc=plugins.py:37  
2026-03-13T02:44:00.437489Z [info      ] setup plugin alembic.autogenerate.defaul  
ts [alembic.runtime.plugins] loc=plugins.py:37  
2026-03-13T02:44:00.437576Z [info      ] setup plugin alembic.autogenerate.commen  
ts [alembic.runtime.plugins] loc=plugins.py:37  
DB: sqlite:///home/hadoop/airflow/airflow.db  
Performing upgrade to the metadata database sqlite:///home/hadoop/airflow/airfl  
ow.db  
2026-03-13T02:44:01.359633Z [info      ] Context impl SQLiteImpl. [alembic.  
runtime.migration] loc=migration.py:210  
2026-03-13T02:44:01.359818Z [info      ] Will assume non-transactional DDL. [alem
```

Starting the Apache Airflow Standalone Environment

```
hadoop@BDTT:~$ python3 -m airflow standalone
2026-03-13T02:45:50.429806Z [info      ] setup plugin alembic.autogenerate.schema
s [alembic.runtime.plugins] loc=plugins.py:37
2026-03-13T02:45:50.429963Z [info      ] setup plugin alembic.autogenerate.tables
[alembic.runtime.plugins] loc=plugins.py:37
2026-03-13T02:45:50.430060Z [info      ] setup plugin alembic.autogenerate.types
[alembic.runtime.plugins] loc=plugins.py:37
2026-03-13T02:45:50.430147Z [info      ] setup plugin alembic.autogenerate.constr
aints [alembic.runtime.plugins] loc=plugins.py:37
2026-03-13T02:45:50.430238Z [info      ] setup plugin alembic.autogenerate.defaul
ts [alembic.runtime.plugins] loc=plugins.py:37
2026-03-13T02:45:50.430319Z [info      ] setup plugin alembic.autogenerate.commen
ts [alembic.runtime.plugins] loc=plugins.py:37
    | Starting Airflow Standalone
    | Password for user 'admin': 7UvawUhpQNKwHBkF
    | Checking database is initialized
```

Accessing the Apache Airflow Web Interface (localhost:8080)

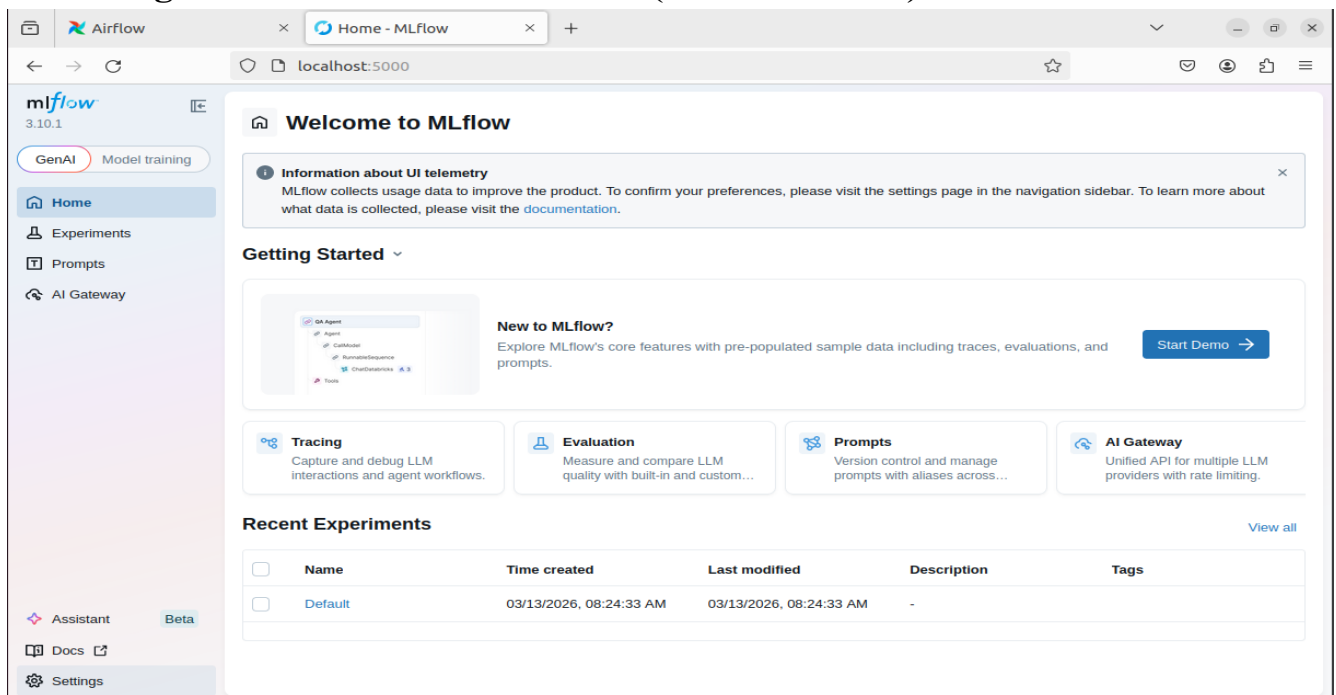
The screenshot shows the Apache Airflow Web Interface running on localhost:8080. The interface has a sidebar with navigation links: Home, Dags, Assets, Browse, Admin, Docs, and User. The main content area displays the following sections:

- Welcome**: The main heading of the interface.
- Stats**: A row of three buttons showing the status of DAGs: Failed Dags (0), Running Dags (0), and Active Dags (0).
- First 10 favorite Dags**: A section for favorite DAGs, currently showing "No favorites yet. Click the star icon next to a Dag in the list to add it to your favorites."
- Health**: A section showing the status of various components: MetaDatabase, Scheduler, Triggerer, and Dag Processor, all marked as "OK". A "Pool Slots" bar shows 128 slots available.
- History**: A section for viewing DAG run history, with a filter for "Last 24 Hours" and a date range from 2026-03-12 08:20:10 to 2026-03-13 08:20:10.
- Dag Runs**: A table showing the status of DAG runs, with columns for Queued, Running, Success, and Failed, all showing 0% completion.
- Asset Events**: A section for viewing asset events, currently showing "No Asset Events found."

Starting the MLflow Tracking Server on Port 5000

```
hadoop@BDTT:~$ mlflow ui --host 0.0.0.0 --port 5000
Backend store URI not provided. Using sqlite:///mlflow.db
Registry store URI not provided. Using backend store URI.
2026/03/13 08:24:31 INFO mlflow.store.db.utils: Creating initial MLflow database
tables...
2026/03/13 08:24:31 INFO mlflow.store.db.utils: Updating database tables
[MLflow] Security middleware enabled with default settings (localhost-only). To
allow connections from other hosts, use --host 0.0.0.0 and configure --allowed-h
osts and --cors-allowed-origins.
INFO:      Uvicorn running on http://0.0.0.0:5000 (Press CTRL+C to quit)
INFO:      Started parent process [61378]
2026/03/13 08:24:38 INFO mlflow.server.jobs.utils: Starting huey consumer for jo
> function optimize_prompts
2026/03/13 08:24:38 INFO mlflow.server.jobs.utils: Starting huey consumer for jo
> function run_online_session_scorer
2026/03/13 08:24:38 INFO mlflow.server.jobs.utils: Starting huey consumer for jo
> function run_online_trace_scorer
2026/03/13 08:24:38 INFO mlflow.server.jobs.utils: Starting huey consumer for jo
> function invoke_scorer
2026/03/13 08:24:38 INFO mlflow.server.jobs.utils: Starting dedicated Huey consu
mer for periodic tasks
```

Accessing the MLflow Web Interface (localhost:5000)



Creating the DAGs Directory for Airflow Pipelines

```
hadoop@BDTT: ~/airflow/dags
hadoop@BDTT:~$ mkdir -p ~/airflow/dags
hadoop@BDTT:~$ cd ~/airflow/dags
hadoop@BDTT:~/airflow/dags$ ls
hadoop@BDTT:~/airflow/dags$
```

Creating the Iris Classifier DAG Python File in the Airflow DAGs Directory

```
hadoop@BDTT: ~/airflow/dags
hadoop@BDTT:~$ mkdir -p ~/airflow/dags
hadoop@BDTT:~$ cd ~/airflow/dags
hadoop@BDTT:~/airflow/dags$ ls
hadoop@BDTT:~/airflow/dags$ nano iris_classifier_dag.py
hadoop@BDTT:~/airflow/dags$ ls
iris_classifier_dag.py
hadoop@BDTT:~/airflow/dags$
```

Writing the Airflow DAG Code to Implement the Automated Iris Classifier Pipeline

```
GNU nano 6.2 iris_classifier_dag.py
from airflow import DAG
from airflow.decorators import task
from datetime import datetime
import mlflow
import mlflow.sklearn

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

C_VALUES = [0.01, 1.0, 10.0]

with DAG(
    dag_id="iris_classifier_pipeline",
    start_date=datetime(2024, 1, 1),
    schedule=None,
    catchup=False,
) as dag:
```

Verifying the Created DAG Using the Airflow DAGs List Command

example_xcom_args	/home/hadoop/.local/lib/python3.10/site-packages/airflow/example_dags/example_xcomargs.py	airflow	True	example_dags	None
example_xcom_args_with_operators	/home/hadoop/.local/lib/python3.10/site-packages/airflow/example_dags/example_xcomargs.py	airflow	True	example_dags	None
iris_classifier_pipeline	/home/hadoop/airflow/dags/iris_classifier_dag.py	airflow	True	dags-folder	None
latest_only	/home/hadoop/.local/lib/python3.10/site-packages/airflow/example_dags/standard/example_latest_only.py	airflow	True	example_dags	None
latest_only_with_trigger	/home/hadoop/.local/lib/python3.10/site-packages/airflow/example_dags/standard/example_latest_only.py	airflow	True	example_dags	None

Triggering the Iris Classifier Pipeline DAG in the Airflow Web Interface

Trigger Dag - iris_classifier_pipeline

Single Run

Trigger a single run of this Dag

☒

Backfill

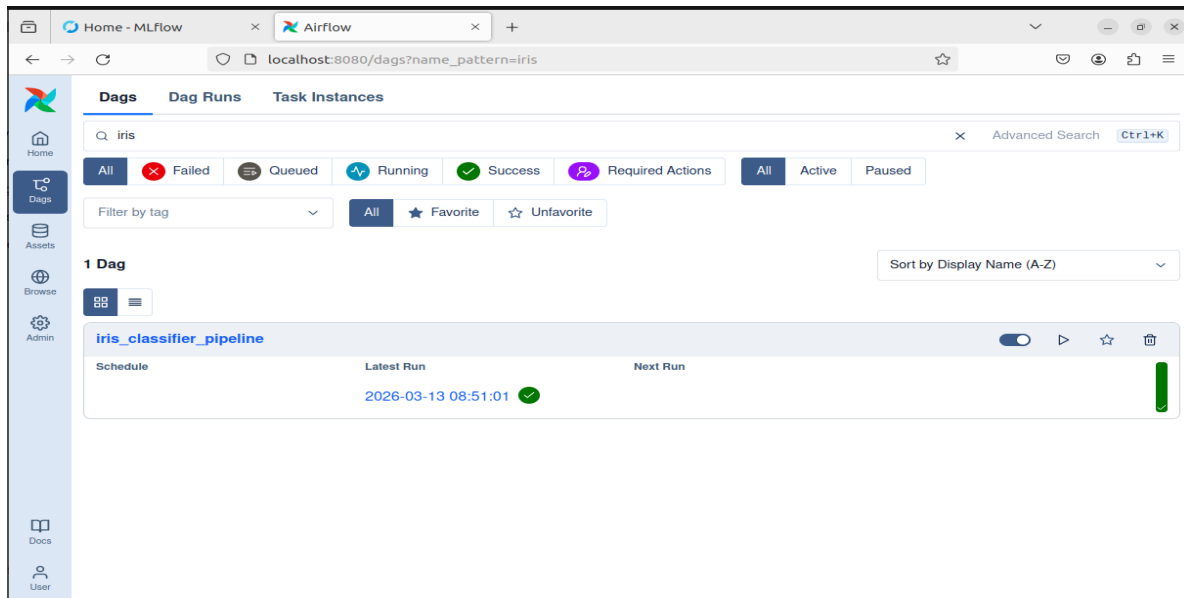
Run this Dag for a range of dates

☐

Advanced Options

▶ Trigger

Verifying the Successful Execution of the Iris Classifier DAG in Airflow



The screenshot shows the Airflow web interface at localhost:8080/dags?name_pattern=iris. The 'Dags' tab is selected, and a search for 'iris' has been performed. The results show one DAG, 'iris_classifier_pipeline', which is in a 'Success' state. The latest run is scheduled for 2026-03-13 08:51:01. The interface includes a sidebar with navigation links (Home, Dags, Assets, Browse, Admin) and a top navigation bar with tabs for Dags, Dag Runs, and Task Instances.

Dags | Dag Runs | Task Instances

Search: iris

Filter by tag: All | Failed | Queued | Running | Success | Required Actions | Active | Paused

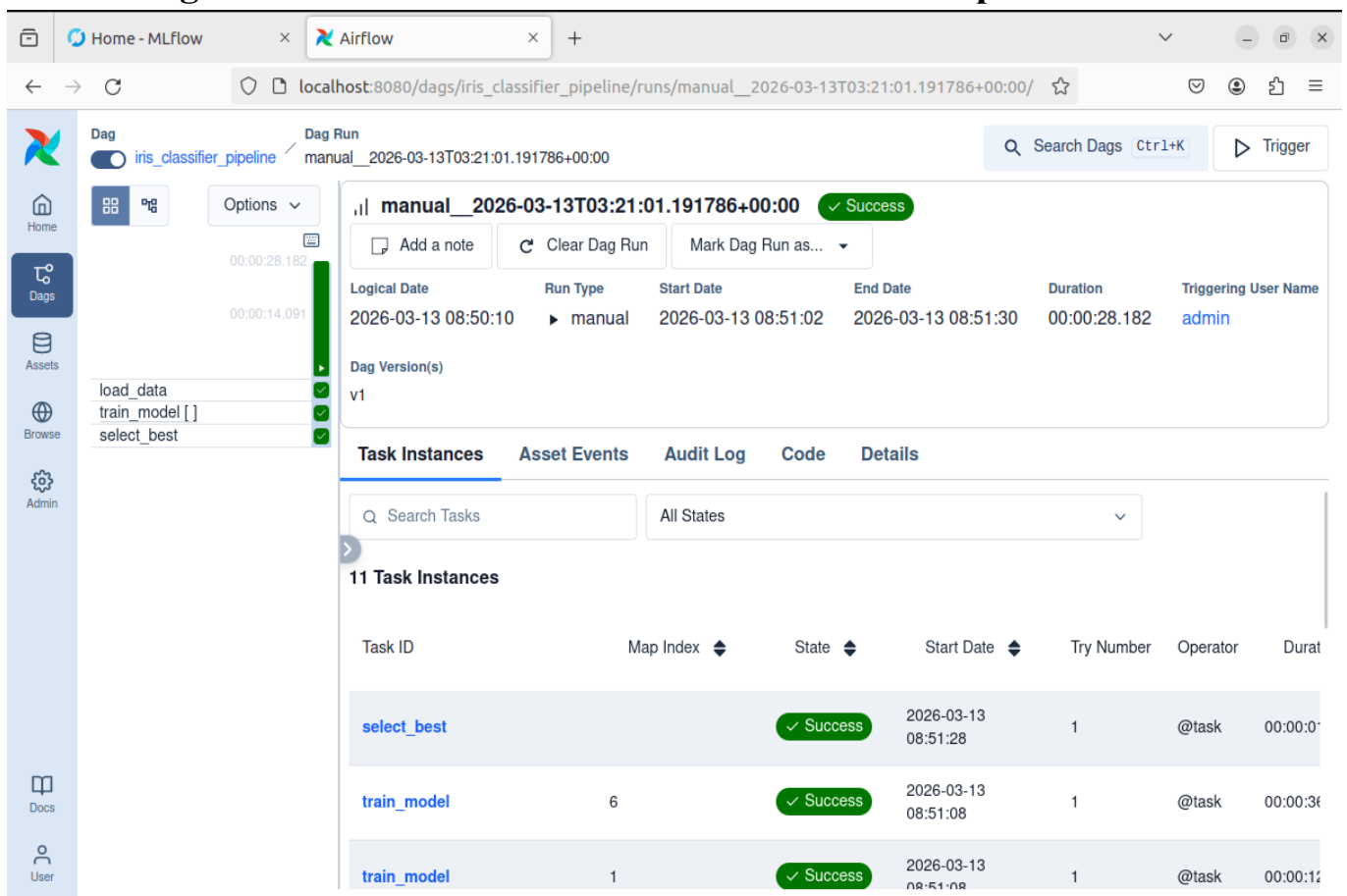
Sort by Display Name (A-Z)

1 Dag

iris_classifier_pipeline

Schedule: 2026-03-13 08:51:01

Monitoring the Execution of Tasks in the Iris Classifier Pipeline DAG



The screenshot shows the Airflow web interface at localhost:8080/dags/iris_classifier_pipeline/runs/manual_2026-03-13T03:21:01.191786+00:00/. The 'Task Instances' tab is selected, and a search for 'manual_2026-03-13T03:21:01.191786+00:00' has been performed. The results show 11 task instances, all in a 'Success' state. The latest run is scheduled for 2026-03-13 08:51:01. The interface includes a sidebar with navigation links (Home, Dags, Assets, Browse, Admin) and a top navigation bar with tabs for Dags, Dag Runs, and Task Instances.

Dag | **Dag Run**

iris_classifier_pipeline | manual_2026-03-13T03:21:01.191786+00:00

Search: manual_2026-03-13T03:21:01.191786+00:00

Trigger

manual_2026-03-13T03:21:01.191786+00:00 (Success)

Logical Date: 2026-03-13 08:50:10 | Run Type: manual | Start Date: 2026-03-13 08:51:02 | End Date: 2026-03-13 08:51:30 | Duration: 00:00:28.182 | Triggering User Name: admin

Dag Version(s): v1

Task Instances | Asset Events | Audit Log | Code | Details

Search Tasks: All States

11 Task Instances

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration
select_best		Success	2026-03-13 08:51:28	1	@task	00:00:00
train_model	6	Success	2026-03-13 08:51:08	1	@task	00:00:36
train_model	1	Success	2026-03-13 08:51:08	1	@task	00:00:12

Viewing the MLflow Experiment Runs Logged During Model Training

The screenshot displays the MLflow web interface at localhost:5000. The left sidebar shows the navigation menu with options like GenAI, Model training, iris_classifier, Runs, Models, Traces, and Model registry. The main area is titled 'Experiments > iris_classifier > Runs'. It features a search bar with the query 'metrics.rmse < 1 and params.model = "tree"', a 'Time created' dropdown, and buttons for 'State: Active', 'Datasets', 'Sort: Created', 'Columns', and 'Group by'. A table lists 9 runs, each with a checkbox, Run Name, Created time, Dataset, Duration, Source, and Models. The runs are: fortunate-kite-540, calm-hawk-46, defiant-ape-224, unique-boar-705, unequaled-croc-628, exultant-slug-346, lyrical-snail-936, fun-snipe-229, and fearless-stoat-289. All runs are marked as successful and sourced from 'airflow'.

Run Name	Created	Dataset	Duration	Source	Models
fortunate-kite-540	8 minutes ago	-	407ms	airflow	-
calm-hawk-46	8 minutes ago	-	272ms	airflow	-
defiant-ape-224	8 minutes ago	-	0.9s	airflow	-
unique-boar-705	9 minutes ago	-	208ms	airflow	-
unequaled-croc-628	9 minutes ago	-	13.1s	airflow	-
exultant-slug-346	9 minutes ago	-	13.4s	airflow	-
lyrical-snail-936	9 minutes ago	-	12.9s	airflow	-
fun-snipe-229	9 minutes ago	-	12.8s	airflow	-
fearless-stoat-289	9 minutes ago	-	1.1s	airflow	-

Visualizing the Iris Classifier Pipeline Workflow in Airflow Graph View

The screenshot shows the Airflow web interface at localhost:8080. The left sidebar includes navigation options like Home, Dags, Assets, Browse, Admin, Docs, and User. The main area displays the 'iris_classifier_pipeline' DAG. The graph shows three tasks: 'load_data @task', 'train_model [9] @task', and 'select_best @task', all of which are marked as successful. The right sidebar provides details for the selected DAG run, including its status (Success), Logical Date (2026-03-13 08:50:10), Start Date (2026-03-13 08:51:02), End Date (2026-03-13 08:51:30), Duration (00:00:28.182), and Triggering User Name (admin). It also shows '11 Task Instances' and a 'Map Index'.

The screenshot displays the Apache Airflow web interface. The top navigation bar shows tabs for 'Dag', 'Dag Run', and 'Task'. The 'Task' tab is active, showing the task 'select_best' with a green 'Success' status. Below the task name is a table with columns: Operator, Start Date, End Date, Duration, and Dag Version. The task details show it was executed by '@task' on 2026-03-13 at 08:51:28, with a duration of 00:00:01.879. The 'Logs' tab is selected, showing a log message source details section with a list of log entries. The log entries include information about DAG bundles, task state transitions (QUEUED to RUNNING), and task completion.

Operator	Start Date	End Date	Duration	Dag Version
@task	2026-03-13 08:51:28	2026-03-13 08:51:29	00:00:01.879	v1

Logs

All Log Levels

All Sources

Log message source details sources=["/home/hadoop/airflow/logs/dag_id=iris_classifier_pipeline/run_id=manual__2026-03-13T03:21:01.191786+00:00/"]

```

2 [2026-03-13 08:51:28] INFO - DAG bundles loaded: dags-folder, example_dags
3 [2026-03-13 08:51:28] INFO - Filling up the DagBag from /home/hadoop/airflow/dags/iris_classifier_dag.py
4 [2026-03-13 08:51:29] WARNING - The 'airflow.decorators.task' attribute is deprecated. Please use 'airflow.sdk.task'
5 [2026-03-13 08:51:29] INFO - Stats instance was created in PID 27801 but accessed in PID 124347. Re-initializing.
6 [2026-03-13 08:51:29] INFO - Task instance is in running state
7 [2026-03-13 08:51:29] INFO - Previous state of the Task instance: TaskInstanceState.QUEUED
8 [2026-03-13 08:51:29] INFO - Current task name:select_best
9 [2026-03-13 08:51:29] INFO - Dag name:iris_classifier_pipeline
10 [2026-03-13 08:51:29] INFO - Champion Run ID: f3600ec71dc446d5bb609b925c980d3
11 [2026-03-13 08:51:29] INFO - Champion Accuracy: 1.0
12 [2026-03-13 08:51:29] INFO - Done. Returned value was: None
13 [2026-03-13 08:51:29] INFO - Task instance in success state
14 [2026-03-13 08:51:29] INFO - Previous state of the Task instance: TaskInstanceState.RUNNING
15 [2026-03-13 08:51:29] INFO - Task operator:<Task(_PythonDecoratedOperator): select_best>
  
```